DATABASE SYSTEMS ARCHITECTURE DOCUMENT

# Banking Database System (BankDB)

Relational Schema Specification & Oracle APEX Implementation Guide

| TARGET RDBMS | DOCUMENT VERSION | SCHEMA OWNER | STATUS |
|---|---|---|---|
| **Oracle Database / APEX** | **1.0.0** | **Student Workspace** | **Production Ready** |

## 1. EXECUTIVE SUMMARY

The Banking Database System (BankDB) is built for Oracle APEX to manage basic everyday banking activities. It securely organizes information for customers, bank accounts, transaction histories, loans, and debit or credit cards. Using clear table relationships and strict database rules, it ensures all bank records stay accurate and organized.

## 2. ENTITIES & DATA DICTIONARY

Below are the column level definitions, data types, and integrity constraints for the database entities.

### 2.1 Table: CUSTOMERS

| Column Name | Data Type | Key Type | Constraints | Description |
|---|---|---|---|---|
| `customer_id` | `NUMBER` | PK | `NOT NULL` | Unique system identification number |
| `first_name` | `VARCHAR2(50)` | — | `NOT NULL` | Customer's primary given name |
| `last_name` | `VARCHAR2(50)` | — | `NOT NULL` | Customer's surname / family name |
| `email` | `VARCHAR2(100)` | — | `UNIQUE, NOT NULL` | Contact email address |
| `phone_number` | `VARCHAR2(20)` | — | Optional | Primary telephone contact number |
| `national_id` | `VARCHAR2(30)` | — | `UNIQUE, NOT NULL` | Government tax or identity number |

### 2.2 Table: ACCOUNTS

| Column Name | Data Type | Key Type | Constraints | Description |
|---|---|---|---|---|
| `account_id` | `NUMBER` | PK | `NOT NULL` | Internal account record identifier |
| `customer_id` | `NUMBER` | FK | `NOT NULL` | Refers to `CUSTOMERS(customer_id)` |
| `account_number` | `VARCHAR2(20)` | — | `UNIQUE, NOT NULL` | Public account number format |
| `account_type` | `VARCHAR2(20)` | — | `NOT NULL` | Type: CHECKING, SAVINGS, etc. |
| `balance` | `NUMBER(15,2)` | — | `DEFAULT 0.00` | Current monetary ledger balance |

### 2.3 Table: TRANSACTIONS

| Column Name | Data Type | Key Type | Constraints | Description |
|---|---|---|---|---|
| transaction_id | NUMBER | PK | NOT NULL | Unique ledger record entry ID |
| account_id | NUMBER | FK | NOT NULL | Refers to ACCOUNTS(account_id) |
| transaction_type | VARCHAR2(20) | — | NOT NULL | DEPOSIT, WITHDRAWAL, TRANSFER |
| amount | NUMBER(15,2) | — | NOT NULL | Monetary transaction value |
| transaction_date | DATE | — | DEFAULT SYSDATE | Execution date & timestamp |

### 2.4 Table: LOANS

| Column | Type | Key |
|---|---|---|
| loan_id | NUMBER | PK |
| customer_id | NUMBER | FK |
| loan_type | VARCHAR2(30) | — |
| amount | NUMBER(15,2) | — |
| interest_rate | NUMBER(5,2) | — |

### 2.5 Table: CARDS

| Column | Type | Key |
|---|---|---|
| card_id | NUMBER | PK |
| account_id | NUMBER | FK |
| card_number | VARCHAR2(16) | UNIQUE |
| card_type | VARCHAR2(10) | — |
| expiry_date | DATE | — |

## 3. ENTITY RELATIONSHIPS & BUSINESS RULES

- **Customer to Accounts (1 : N):** A customer can own multiple bank accounts (Checking, Savings), but each account is owned by exactly one customer.
- **Account to Transactions (1 : N):** An account maintains a historical log of multiple monetary transactions. Each transaction links back to one parent account.
- **Customer to Loans (1 : N):** A customer can apply for and hold multiple active credit products or loans.
- **Account to Cards (1 : N):** One or more debit or credit cards can be linked directly to an existing bank account.